

FPGA CODEC comparisons										James Brakefield										Advances in RNA Appendix A									
directory name: BM/NS H/_ generic format & size	memory format	ALU bits	ALU exp	ALU signifi cand	zero NaR stat	ALU roun ding	mem bits	max exp bits	fract size max	fract size min	total binad es	decad es / 2	mem BMF?	mem HUB?	directi on	FPGA device	LUT cnt	DSP cnt	delay ns	fpga tool	fpga optimization	run date	test bench	valid ated	RTL author VHDL-2008	comments			
	The four memory format types																												
NSHid_takum16	Takum	24	9	13	1		16	9	11	4	512	77.1	NFSF	HUB	load	zynq uscale+										Laslo Hunhold	standard linear takum		
NSHid_bposit16	Posit sB=5	24	9	13	1		16	8	8	0	400	60.2	NFSF	HUB	load	zynq uscale+										Florent de Dinec	Sb field always present, e.g. limited regime field		
NSHid_wmesa16	wide Mesa	24	9	13	1		16	6	9	0	352	53.0	NFSF	HUB	load	zynq uscale+										Jim Brakefield	wide mesa		
NSHid_pedestal16	pedestal	24	9	13	1		16	8	7	7	254	38.2	NFSF	HUB	load	zynq uscale+										Jim Brakefield	min implementation of Gustafson criteria		
NSHid_nmesa16	narrow Mesa	24	9	13	1		16	6	9	0	52	7.8	NFSF	HUB	load	zynq uscale+										Jim Brakefield	P3109 mesa		
	Pedestal with HUB encoding for zero, NaR, +/- 1.0, no sub or super normals																										for comparison to Bfloat16		
NS_id_pedestal16	pedestal8_16	24	9	13	1	RTE	16	8	7	7	254	38.2	NFSF		load	zynq uscale+	3		0.940	vivado 25.2	area remap	06/04/26			Jim Brakefield	zero_NaR the only logic			
NSHid_pedestal16	pedestal8_16	24	9	13	1	ITO	16	8	7	7	254	38.2	NFSF	HUB	load	zynq uscale+	3		0.810	vivado 25.2	area remap	06/04/26			Jim Brakefield	decpde zero, NaR and +/- 1.0			
NS_st_pedestal16	pedestal8_16	24	9	13	1	RTE	16	8	7	7	254	38.2	NFSF		store	zynq uscale+	21		1.120	vivado 25.2	area remap	06/04/26			Jim Brakefield	round to nearest, ties to even			
NSHst_pedestal16	pedestal8_16	24	9	13	1	ITO	16	8	7	7	254	38.2	NFSF	HUB	store	zynq uscale+	8		0.750	vivado 25.2	area remap	05/28/26			Jim Brakefield	encode zero, NaR and +/- 1.0			
	Wide taper Mesa with HUB encoding for zero, NaR, +/- 1.0																										wide mesa, nearly Takum exponent range		
BM_id_wMesa6_16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	BMF		load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
BMHid_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_id_wMesa6-16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	NFSF		load	zynq uscale+	21	0	1.355	vivado 26.1	area remap	08/25/26			Jim Brakefield				
NSHid_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	NFSF	HUB	load	zynq uscale+	16	0	1.341	vivado 26.1	area remap	08/25/26			Jim Brakefield				
BM_st_wMesa6-16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	BMF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
BMHst_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_st_wMesa6-16	wMesa6-16	24	9	13	1	RTE	16	6	9	0	352	53.0	NFSF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NSHst_wMesa6-16	wMesa6-16	24	9	13	1	ITO	16	6	9	0	352	53.0	NFSF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
	Narrow taper Mesa with HUB encoding for zero, NaR, +/- 1.0																										for comparison to IEEE Float16		
BM_id_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	BMF		load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
BMHid_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_id_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	NFSF		load	zynq uscale+	34	0	0.956	vivado 26.1	area remap	08/22/26			Jim Brakefield				
NSHid_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	NFSF	HUB	load	zynq uscale+	24	0	1.404	vivado 26.1	area remap	08/22/26			Jim Brakefield				
BM_st_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	BMF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
BMHst_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_st_nMesa5_16	nMesa5_16	24	9	13	1	RTE	16	5	10	0	52	7.8	NFSF		store	zynq uscale+	242	0	4.850	vivado 26.1	area remap				Jim Brakefield	incorrect round ties to even			
NSHst_nMesa5_16	nMesa5_16	24	9	13	1	ITO	16	5	10	0	52	7.8	NFSF	HUB	store	zynq uscale+	32	0	1.362	vivado 26.1	area remap	09/04/26			Jim Brakefield				
	Linear Takum with HUB encoding for zero, NaR, +/- 1.0																										linear and logarithmic varinats defined		
BM_id_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	BMF		load	KntexUltrSca+	39		3.652	vivado 24.1	default				Laslo Hunhold	from paper's TeX:Dsgn & Imp takum arithmetic codec			
BMHid_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_id_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	NFSF		load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NSHid_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	NFSF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
BM_st_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	BMF		store	KntexUltrSca+	66		4.180	vivado 24.1	default				Laslo Hunhold	from paper's TeX:Dsgn & Imp takum arithmetic codec			
BMHst_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	BMF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_st_Takum	Takum	24	9	13	1	RTE	16	9	11	4	512	77.1	NFSF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NSHst_Takum	Takum	24	9	13	1	ITO	16	9	11	4	512	77.1	NFSF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
	Wide Posit (eS=5) with HUB encoding of zero, NaR, +/- 1.0																										HUB and logarithmic variants studied		
BM_id_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	BMF		load	KntexUltrSca+	57		3.911	vivado 24.1	default				Florent de Dinec	from paper TeX:Dsgn & Imp takum arithmetic codec FioPoCo			
BM_st_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	BMF		store	KntexUltrSca+	56		4.349	vivado 24.1	default				Florent de Dinec	from paper TeX:Dsgn & Imp takum arithmetic codec FioPoCo			
BMHid_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	BMF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
BMHst_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	BMF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_id_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	NFSF		load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NS_st_Posit_eS5	Posit sB=5	24	9	13	1	RTE	16	9	11	4	400	60.2	NFSF		store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NSHid_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	NFSF	HUB	load	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
NSHst_Posit_eS5	Posit sB=5	24	9	13	1	ITO	16	9	11	4	400	60.2	NFSF	HUB	store	zynq uscale+				vivado 26.1	area remap				Jim Brakefield				
	To Establish Suitable Context for the CODEC results																										codecs used on memory load/store, not within ALU?		
	add/subtract	24	9	13	1	ITO										zynq uscale+	300-400				vivado 26.1	area remap				AMD IP	pipelined, no speed optimization, round ties to even?		
f12_e9_mult_zu3	multiply	24	9	13	1	ITO	NA	10	13	NA	512	77.1	NA	NA	NA	zynq uscale+	43	1	2.700	vivado 26.1	area remap	07/27/26			Jim Brakefield	saturation, extra exp bit, hidden fract bit			
f12_e9_mult_zu3	multiply	24	9	13	1	ITO	NA	10	13	NA	512	77.1	NA	NA	NA	zynq uscale+	34	1	2.920	vivado 26.1	area remap	07/28/26			Jim Brakefield	ibid. clk of 4.0ns instead of 3.0ns			
	divide combinatorial	24	9	13	1	ITO										zynq uscale+				vivado 26.1	area remap				AMD IP				
	divide via reciprocal table	24	9	13	1	ITO										zynq uscale+				vivado 26.1	area remap				Jim Brakefield	uses block RAM for reciprical lookup			

Round Inexact To Odd (IEEE P754 Round to one) **ITO**

Round ties to even **RTE**

NFSF: No Fraction Shifter Format **NS**

BMF: Binary Monotonic Format **BM**

HUB: Half Unit Bias AKA Implicit half bit set format **H**

Binade: Approximate binary mem exp range: total binary exponent range

For this study, all binades have atleast one entry

1/2 Decade: Approximate half decimal range, e.g. +/- total decimal exponent range

On HUB load implicit HUB bit is inserted into the ALU fraction

On store and non-HUB the ALU fraction is rounded ties to even

In general on load and store the resulting value is true to the source value, rounded as appropriate

Timing results via enclosing combinatorial logic with input and output registers

The pedestal AKA flat roof format supports zero, NaR and saturation and no sub-normals

Directory naming: BM/NS, H/_ LD/ST, _name-size

Mesa 6-48: six bit exponent with 48 of the 64 exponents used on the tapers (sub and super-normals)

Fraction size of zero means only the hidden leading one bit present

Saturation can occur on memory stores due to exponent size reduction or if round ties to even increases the exponent

Four somewhat affordable FPGA boards with most modern chip fabrication are:

TerasiC DE23-Lite (Altera Agilex 3)	Altera part #: A3C2135B818AE75	\$182	X
Real Digital's AUP-ZU3 (AMD Zynq UltraScale+)	xilinx part #: XCZU3EG-2SFVC784E	\$309	X
realdigital.org Boolean Board (xilinx spartan-7)	xilinx part #: XC7S50 CSGA324ABX	\$84	X
Adiuvo Explorer board (AMD Artix ultrascale+)	xilinx part #: XCAU7P	\$99	